

# Parviz Narimani

Lecturer at KNTU
Founder of Alvyn AI Studio


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## Profile

I am a multidisciplinary AI and engineering researcher with a strong foundation in mechanical engineering, civil engineering, and applied machine learning. My work centers on integrating data-driven modeling, intelligent systems, and engineering design, enabling me to contribute to diverse domains such as advanced manufacturing, materials engineering, robotics, and sustainable infrastructure. This cross-disciplinary expertise has resulted in a growing research portfolio, including several published papers, 80 citations, an h-index of 4, and an i10-index of 3, demonstrating both productivity and impact across engineering and AI-driven applications.

My multidisciplinary trajectory began with hands-on engineering and robotics work, including my role as Co-Founder and CTO of Tart Robotics, where I led the development of educational robotics systems and contributed to international product deployment. I later expanded my strategic and managerial capabilities through a Doctor of Business Administration (DBA) at the University of Tehran, which strengthened my expertise in AI-driven organizational transformation and strategic management. Today, as a University Lecturer at K.N. Toosi University of Technology, I teach Artificial Intelligence for Future Business Managers, helping the next generation of leaders adopt data-driven decision-making and integrate AI strategically into business contexts.

## Education

### Doctor of Business Administration – Major in General Strategy
*2023 - 2025*

***University of Tehran, Tehran, Iran***

***Project Title:*** *"*Strategy implementation of Alvyn AI Studio as a new born start-up in the field of artificial intelligence"

*Supervisor: Dr. Hoeesin Safari*

### Master of Science in Mechanical Engineering – Design and applied mechanics
*2015 - 2018*

***University of Tehran, Tehran, Iran***

- *Ranked 201-250th in QS World University Rankings by Subject 2022, Mechanical Engineering*

***Thesis Title:*** *"*Design and implementation of visual-servoing controller for a laparoscopic camera holding robot"

*Supervisor: Dr. Farshid Najafi*

### Bachelor of Science in Mechanical Engineering
*2010- 2014*

***University of Tehran, Tehran, Iran***

- *Ranked 201-250th in QS World University Rankings by Subject 2022, Mechanical Engineering*

***Final Project:*** *"*Design and implementations of mini size quadrotor*"*

*Supervisor: Dr. Farshid Najafi*

### National Organization for Development of Exceptional Talents (SAMPAD)
*2003- 2010*

***Shahid Soltani, Karaj, Iran***
***Middle and High School***
***Major:*** Mathematics

## Publications (International Journals and Conferences)

### Published Papers

Characterisation of structural performance and sustainability of hot mix asphalt mixtures containing RAP using Machine Learning — *2026*
*Australian Journal of Civil Engineering*
https://doi.org/10.1080/14488353.2025.2609053
*A. Jamshidi, P. Narimani, R. Haque and MRM. Hasan*

Analytical and Machine-Learning Framework for Predicting the Impact of LPBF Parameters on the Hardness of Inconel™ 718 — *2025*
*Journal of Materials Engineering and Performance*
https://doi.org/10.21203/rs.3.rs-7648647/v1
*M.D. Abyaneh, R. Sesana, M. S. Javadi, P. Narimani, M. Crachi, A. Caraviello*

Enhancing Grinding Efficiency in Aluminum Alloys: An Ensemble-Stacking and Single Machine Learning Framework for Predicting Surface Roughness with SHAP-based interpretability — *2025*
*The International Journal of Advanced Manufacturing Technology*
https://doi.org/10.21203/rs.3.rs-7647149/v1
*M.D. Abyaneh, M.S. Javadi, P. Narimani, M. Golabchi, R. Sesana, M. Hadad*

Machine Learning Approaches for Predicting the Rheological Characteristics of Malaysian Asphalt Binders Containing Synthetic Wax — *2024*
*12th MRC & Exhibition 2024 and 5th IRF Global Asia-Pacific Regional Congress*
*A. Jamshidi, P. Narimani, M.M. Hasan, G. White and M.O.Hamzah*

Predicting Surface Roughness and Grinding Forces in UNS S34700 Steel Grinding: A Machine Learning and Genetic Algorithm Approach to Coolant Effects — *2024*
*Physchem (MDPI), journal on science and technology in physical chemistry*
https://doi.org/10.3390/physchem4040035
*M.D. Abyaneh, P. Narimani, M.S. Javadi, M. Golabchi, S. Attarshargi and M.J. Hadad*

Digitalization of Analysis of a Concrete Block Layer Using Machine Learning as a Sustainable Approach — *2024*
*Sustainability in Civil and Infrastructure Engineering [MDPI]*
https://doi.org/10.3390/su16177591
*P. Narimani, M.D. Abyaneh, M. Golabchi, B. Golchin, R. Haque, A. Jamshidi*

Exploring New Parameters to Advance Surface Roughness Prediction in Grinding Processes for the Enhancement of Automated Machining — *2024*
*Journal of Manufacturing and Materials Processing [MDPI]*
https://doi.org/10.3390/jmmp8010041
*M.J. Hadad, S. Attarshargi, M.D. Abyaneh, P. Narimani, J. Makarian, A. Saberi and A. Alinaghizadeh*

A novel approach to modeling of environmentally friendly grinding of steel St37 using an advanced artificial neural network — *2023*
*Energy Equipment and Systems*
https://doi.org/10.22059/ees.2023.2000614.1424
*M.D. Abyaneh, P. Narimani, S. Attarshargi and M.J. Hadad*

## Submitted Papers

Machine Learning–Based Characterization and Digitalization of Dynamic Modulus Models for Hot Mix Asphalt — *2026 submitted*
*Journal of Traffic and Transportation Engineering (English Edition)*
*A. Jamshidi, P. Narimani, MRM Hasan*

Cumulative Machine Learning and Digitalization Insights into Grinding Processes with Multi-Alloy and Multi-Cooling Variability — *2026 submitted*
*Advances in Manufacturing*
*P. Narimani, M. D. Abyaneh, H. Hamzeh, R. Sesana, M. Hadad and A. Alinaghizadeh*

Advancing Sustainable Pavement Design: A Data-Driven ML Framework for Predicting the Rheological Performance of Sasobit™-Modified Binders — *2026 submitted*
*Australian Journal of Civil Engineering*
*M. D. Abyaneh, P. Narimani , H. Hamzeh , M. S. Javadi, K. Rahsepas*

## Ongoing Papers

From Visualization to Decision: Introducing MBDF for Scalable Model Comparison in Multidisciplinary Machine Learning — *Mar. 2026 expected*
*P. Narimani, M. D. Abyaneh*

Towards Intelligent Trading Strategies in Binary Options: A Machine-Learning-Empowered Multimodal Analytical Framework for Candlestick Pattern Discrimination and Predictive Trade Inference, Substantiated Through Empirical Real-World Validation — *Apr. 2026 expected*
*P. Narimani, F Firouzi*

## Reviewer (International Journals and Conferences)

| | | |
|---|---|---|
| **Journal of Diabetes & Metabolic Disorders** | Deep learning based Binary Classification of Diabetic Retinopathy Images using Transfer Learning Approach | *2024* |
| **PLOS One** | Predicting Diabetes Mellitus Metabolic Goals and Chronic Complications Transitions—Analysis Based on Natural Language Processing and Machine Learning Models | *2024* |

## Professional Certifications

| | | |
|---|---|---|
| Microsoft® | **AI for Organizational Leaders by Microsoft and LinkedIn** | 2026 |
| Microsoft® | **Career Essentials in Generative AI by Microsoft and LinkedIn** | 2025 |
| LinkedIn® | **Explore a Career in Product Management** | 2023 |
| Aha!® LinkedIn® | **Aha! Product Management Professional Certificate** | 2023 |
| LinkedIn® | **Master Digital Transformation** | 2023 |
| LinkedIn® | **Advance Your Digital Transformation Strategy as a Leader** | 2023 |
| LinkedIn® | **LinkedIn Marketing Labs: Marketing Strategy Certification** | 2023 |

## Professional Experiences

**Business Line Manager** *2025- present*

*Faran® Gold* | ***Website*** | ***Linked In***

- Business Strategy
- Digital Transformation
- Digital Footprint

**University Lecturer | Artificial Intelligence** *2024 - present*

*K. N. Toosi University of Technology | Faculty of General Managements*

- Business Management
- Business Strategy
- Artificial Intelligence
- Generative AI

**Founder** *2024 - present*

*Alvyn AI Studio* | ***Website*** | ***Linked In***

- Business Strategy
- Business Development
- Artificial Intelligence
- Generative AI

**Co-Founder | CTO** *2018 - 2020*

*Tart Robotics®* | ***Linked In***

- Product Management
- Lego® Education
- Robotics
- Mechatronics

**Research Assistant | Robotics Specialist** *2013 - 2020*

*Robome - Robotics Engineering Center | University of Tehran* | ***Linked In***

- Bio-inspired Robotics
- Surgery Robotics
- Educational Robotics
- Embedded System

## Technical Skills

- **Artificial Intelligence**
- **Machine Learning**
- **Robotics**
- **Business Strategy**
- **Product Management**
- **Product Innovation**
- **Lean Foundation**
- **Generative AI**
- **Mechatronics**

## Reference

Dr. Farshid Najafi
Webpage
*Assistant Professor of Professional Practice, Graduate Student Supervisor Faculty of Applied Sciences, Mechatronic Systems Engineering, Simon Fraser University, B.C. Canada*
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Dr. Ali Jamshidi
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*Adjusant Professor, School of Engineering and Science, University of the Sunshine Coast, Sunshine Coast, Queensland, Australia*
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Dr. Raffaella Sesana
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*Associate Professor, Department of Mechanical and Aerospace Engineering (DIMEAS), Politecnico di Torino, Turin, Italy*
raffaella.sesana@polito.it